

Linking network structures and stochastic flow properties: An exploratory Markov-spectral case study in professional football

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Abstract

Understanding team performance in professional football increasingly benefits from network science, which models players as nodes and their interactions as functional links. While Social Network Analysis (SNA) provides valuable structural insights, it often fails to capture the probabilistic nature of possession. This exploratory study introduces an integrated methodological framework that combines static SNA with a Markov-spectral model of ball circulation to analyse the Portuguese National Team across two high-stakes 2025 Nations League finals matches. This approach moves beyond describing *who* is connected to quantifying *how* possession flows through the team network. Match event data were obtained via the Wyscout[®] platform, with all passing actions by Portugal extracted. The framework combined SNA (using uPATO[®]) with Markov-spectral analysis (implemented in MATLAB[®]) to compute indices of passing uncertainty, diffusion speed, navigability, and network robustness. The results suggested that (i) SNA identified key players as crucial hubs for ball circulation at the micro level, while indicating a cohesive team structure with adaptable macro-level properties; and (ii) Markov-spectral quantification showed descriptive between-match variations, with the second match displaying higher Entropy (2.84 vs 2.77 bits/pass), suggestive of greater unpredictability, and a larger Spectral Gap (0.53 vs 0.45), indicative of faster potential diffusion of possession. Overall, this integrated approach demonstrates the feasibility of profiling team coordination through both structural configuration and stochastic flow properties. The Markov-spectral framework complements traditional SNA by providing quantifiable indices related to passing variability, network navigability, and structural cohesion, offering a multi-layered, proof-of-concept toolkit for analysing collective performance.

Keywords

team dynamics, collective behaviour, performance analysis, social network analysis, Markov-spectral modelling, football, Portuguese National Team

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Introduction

Understanding team performance in professional football increasingly benefits from the application of network science^{1,2} which conceptualises players as nodes and their interactions as functional links. This perspective allows researchers to capture the emergent properties of collective behaviour and tactical organisation,^{2,3} identifying key players, team tactics, and patterns of ball circulation,^{1,4-9} moving beyond reductionist analyses of isolated technical or tactical actions. This perspective is aligned with a dynamic systems approach to football, which views teams as complex, adaptive systems where performance emerges from the continuous interaction of players under multiple constraints.^{2,10} As highlighted by Gama et al.,² network analysis also reveals how structural and functional coordination emerges from the patterns of interaction among players, providing metrics that may reflect team adaptability and strategic efficiency.

While Social Network Analysis (SNA) has provided a valuable static snapshot of team structure and key players, most studies remain descriptive and structurally focussed.^{1,2,7,8} Consequently, a methodological gap exists in quantifying the probabilistic nature of possession and its link to functional aspects of team play, such as the uncertainty inherent in passing sequences, network navigability, and systemic robustness.¹¹⁻¹³ Indeed, a recent systematic review of football performance analysis highlighted the need to integrate temporal aspects into network studies to fully understand team behaviour.^{2,3,14,15}

Likewise, while prior research has successfully established the Markov-chain model as a valid approximation for the growth of passing networks over time¹⁶ and to model state transitions between field zones,¹⁷ its potential to describe the relational and flow properties of the player network itself remains largely untapped. Markovian processes naturally accommodate such probabilistic, time-evolving phenomena by modelling sequences of play as state transitions on a network,¹⁸ offering a promising avenue to address this shortfall in analysis of flow properties.

However, a relevant methodological gap persists: while SNA describes *who* is connected and identifies structural hubs,^{1,2,4} and Markov models have been applied either to network growth over time¹⁶ or to state transitions between field zones,¹⁷ no framework has yet systematically combined structural topology with stochastic flow metrics to quantify real-time tactical functionality on the player network itself, such as how quickly possession diffuses (Spectral Gap), how unpredictably it circulates (Entropy Rate), or how efficiently it reaches key players (Mean First-Passage Time). This integration gap limits the ability to move from descriptive structural analysis to explanatory profiling of tactical functionality and efficiency, a challenge recently outlined in the literature.^{2,3,15} Therefore, this study bridges this methodological gap by integrating static

SNA with a dynamic Markov-spectral framework for flow properties to model ball circulation as a stochastic diffusion process on the player network. While previous studies have applied Markov models to football passing networks, for example, to analyse network growth over time¹⁶ or state transitions between field zones,¹⁷ the present framework makes a distinct and novel contribution. Specifically, it provides spectral and entropy indices suggestive of tactical adaptability, network navigability, and systemic robustness that cannot be captured by static SNA metrics alone. Unlike conventional centrality or connectivity measures, the proposed Markov-spectral model yields interpretable dynamic quantities such as the Entropy Rate (uncertainty in pass destination), Spectral Gap (speed of possession diffusion), Kemeny's Constant (network-wide navigability), and Algebraic Connectivity (structural robustness), all derived directly from the player-passing network. This integrated approach is, to our knowledge, the first to systematically combine these two lenses to provide a unified structural and dynamic profile of team behaviour. It allows us to move beyond describing *who* is connected to describing *how* possession flows, how efficiently it navigates the network, and how robust the system is under pressure. This integrated approach is, to our knowledge, the first to systematically combine these two lenses to provide a unified structural-dynamic profile of team behaviour. It allows us to move beyond describing *who* is connected to describing *how* possession flows, how efficiently it navigates the network, and how robust the system is under pressure, questions that static SNA alone cannot answer.

To guide this exploratory investigation, the following research questions (RQs) were formulated: **(RQ1)** How can an integrated SNA and Markov-spectral framework capture both structural and stochastic flow properties of elite football passing networks? **(RQ2)** How do network topology and stochastic flow properties differ between matches under maximum competitive pressure; and **(RQ3)** To what extent do Markov-spectral indices provide novel insights into tactical adaptability, network navigability, and systemic robustness beyond traditional SNA metrics? Correspondingly, it was expected that: **Hypothesis 1 (H1)** SNA will identify stable structural hubs, while Markov-spectral indices will reveal between-match variability; **Hypothesis 2 (H2)**: network cohesion will be maintained despite tactical variations, reflected in increased Entropy and Spectral Gap; and **Hypothesis 3 (H3)** Spectral and Entropy-based metrics will quantify functional properties orthogonal to static centrality measures.

This study employs an exploratory approach, analysing the Portuguese National Team across two key matches of the 2025 Nations League finals, with the primary aim of demonstrating the feasibility of an integrated application of SNA and Markov-spectral modelling in elite football performance. This pilot focus on a championship-winning team under high-stakes

conditions provides a critical test case, ensuring the framework is evaluated where tactical performance demands are maximal.

Materials and methods

Study design and rationale

This research employs a methodological pilot design with the primary objective of developing and providing a proof-of-concept for the novel integrated analytical framework that integrates SNA with Markov-spectral modelling. The study is structured as a proof-of-concept investigation, conducted on a high-performance team in a high-stakes, ecologically valid competitive setting (the 2025 Nations League finals), to demonstrate the framework's feasibility, computational implementation, and capacity to detect between-match variations in network-derived metrics.

Consistent with recent recommendations for exploratory research in sport science,^{19–21} the two-match sample is sufficient to illustrate the application of the integrated framework and generate preliminary observations. This approach aligns with previous exploratory studies that employed small samples (one-two matches) to establish methodological proof-of-concept, to investigate tactical dynamics, or to propose new analytical frameworks.^{11,22–26} Furthermore, the integration of Markov chain models with SNA follows methodological precedent for examining interaction patterns and unpredictability in elite football.^{24–25}

The following primary validation criteria were established for this proof-of-concept: (i) Computational feasibility: can all indices be reliably computed from standard match data? (ii) Internal consistency: do metrics derived from the same underlying adjacency matrix yield interpretable relationships? and (iii) Sensitivity to observable match-to-match variation: do the metrics show differences between matches that align with expected tactical distinctions (e.g. formation change and opponent characteristics)? The two-match sample is sufficient to assess these criteria, as each match provides hundreds of passing events from which stable transition probabilities and network metrics can be estimated. As highlighted in systematic reviews of guidance for exploratory studies,²⁷ there is no universal consensus on sample size requirements for such investigations, and the value of exploratory research derives from factors such as sample uniqueness and data quality rather than sample size alone.¹⁹

Establishing these foundational parameters is essential before extending the methodology to broader ecological contexts and larger-scale datasets.

Sample

As an exploratory study, the analysis focussed on a single team (the Portuguese National Team) during the final stage of the 2025 Nations League, covering only

the semi-final and final. In the semi-final (Match 1), Portugal faced Germany, securing a win with a 2-1 scoreline and registering 45.54% possession. In the final (Match 2), against Spain, the team drew by 2-2 (winning after extra time on penalties – 5-3), with 27.8% possession. This contextual information is provided to support the interpretation of observed network variations, as opponent quality and match context are known to influence passing patterns.^{2,8,28,29} The Portuguese National Team was selected as it won the competition, allowing a detailed examination of both individual and collective performance in a high-stakes environment. Analysing a championship team under peak competitive pressure provides an opportunity to observe network attributes associated with successful performance, a strategy used in prior studies to identify features related to winning play.^{1,30}

Data collection and analysis

Match data were obtained via the Wyscout® platform, which provides validated metrics widely used in sports research.^{31–33} Pappalardo et al.³² reported a global inter-operator agreement rate (IRAR) of 0.70 for Wyscout event data, indicating good reliability. More recently, Cao³³ conducted a specific validation of passing events, confirming high accuracy and consistency in passer and receiver identification across multiple matches. All passing actions executed by Portugal across the semi-final and final were extracted, including passer and receiver identities, ensuring the entire network of on-ball interactions was captured.^{2,4,5}

After extraction, the data were processed through the following steps: (i) filtering all passes involving Portugal players; (ii) coding each event by passer and receiver identity; (iii) aggregating pass frequencies per match to construct directed, weighted adjacency matrices, where each entry a_{ij} represents the total number of passes from player i to player j ; and (iv) exporting these matrices for analysis in uPATO® and MATLAB®. Each directed edge was weighted by pass frequency, quantifying connection strength on the field.

To facilitate interpretation of the network analyses and Markov-spectral modelling, the principal mathematical symbols and nomenclature used throughout the study are summarised in Supplemental Section S1.

All quantities were computed per match from the same adjacency input for internal consistency. In addition to per-match analyses, an aggregated network representing passing behaviour across both matches was constructed by summing pass frequencies from Match 1 and Match 2 prior to computation of all indices.

The datasets were imported into uPATO® for SNA,^{9,26,34} where key SNA metrics were computed. Subsequently, adjacency matrices were exported to MATLAB® for Markov-spectral modelling, yielding Stationary Distributions, Entropy Rates, Spectral Gaps, Kemeny's Constants, Mean First-Passage Times

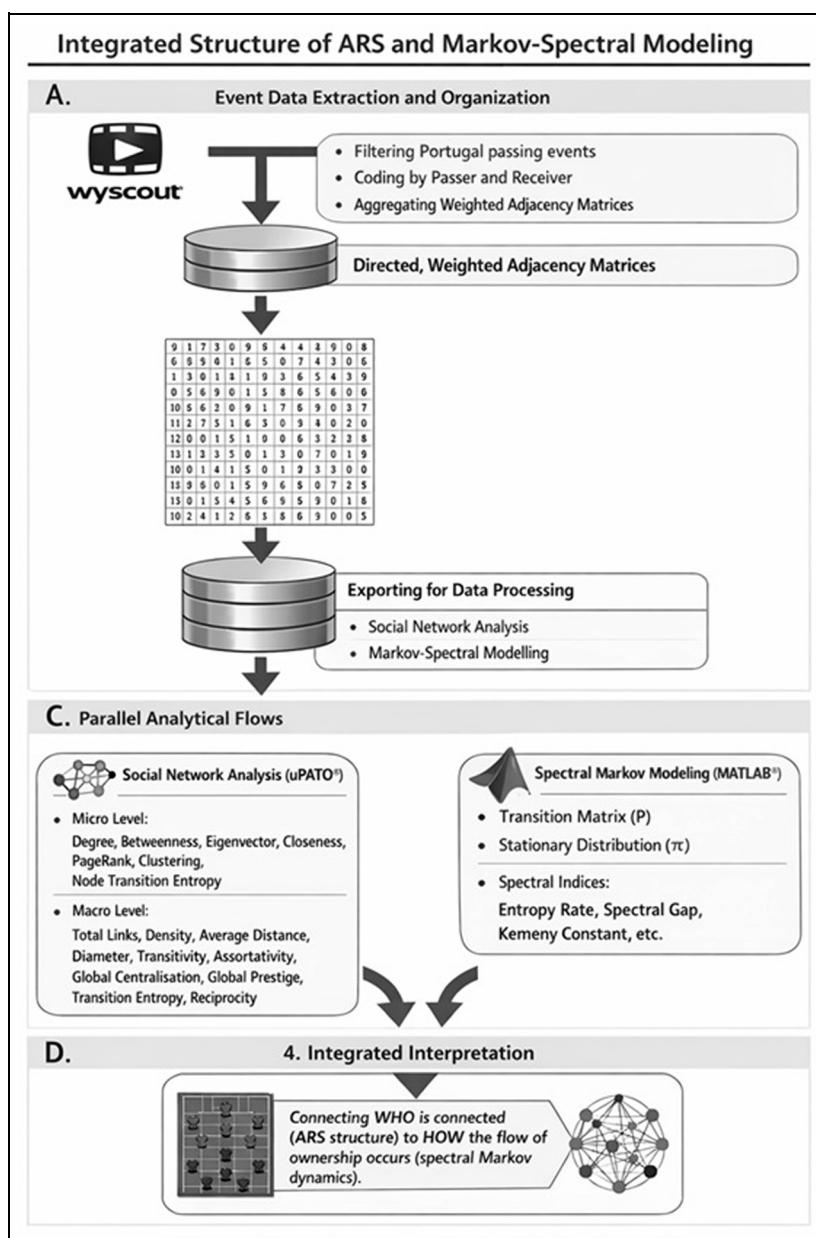


Figure 1. Schematic overview of the integrated analytical framework.

(MFPT), Asymmetry Indices, Algebraic Connectivity, von Neumann Entropy, and Kirchhoff Indices.

The complete analytical pipeline is summarised schematically in Figure 1.

Social network analysis

SNA offers a robust quantitative framework for modelling team sports by representing players as nodes and their interactions as edges within a complex network.^{1,2} This methodology transcends traditional reductionist approaches that focus on isolated events, enabling the examination of emergent individual and collective behaviours, including identification of key influencers, preferred playing zones, tactical organisations, and patterns of collective behaviour.^{3–5,7} Previous research has

demonstrated that cohesive network structures and well-chosen centrality metrics can illuminate performance outcomes. For example, teams with higher connectivity and balanced interaction patterns often achieve greater success.^{30,35}

Within SNA, two complementary families of metrics are distinguished.^{2,3,15} Centrality measures (Degree, Betweenness, Eigenvector, Closeness, and Stress) quantify structural role and involvement level, answering “*who participates most*” and “*who connects whom*.”^{1,4} These reflect the volume and positional importance of contributions to the passing network. Entropy-based measures (Node Transition Entropy and Network Transition Entropy) capture variability and unpredictability in passing choices, answering “*how diversely the ball is distributed*” and “*how difficult it is to anticipate*

the next pass.”^{24,25,36} A player may have high degree centrality by repeatedly passing to the same teammate (low entropy, predictable), while another with moderate degree but high entropy distributes passes more evenly (unpredictable, harder to defend).²⁵ Both dimensions are necessary for a complete tactical profile.^{2,3,15}

In the present study, mathematical models integrated with SNA were applied to provide a comprehensive assessment of both individual player performance (micro-level) and overall team dynamics (macro-level):

1. Micro-level (individual): A comprehensive range of SNA metrics was computed to profile individual player involvement. Degree, Closeness, Betweenness, and Eigenvector centralities were prioritised as indicators of influence and mediation of information flow, consistent with previous football network studies.^{1,2,4–6,30,35} Node Transition Entropy was included to capture variability in passing choices, complementing the centrality metrics.^{24,25} Additionally, other metrics, including Stress, Eccentricity, Degree Prestige, PageRank, Power, Subgraph, Centroid, Proximity Prestige, and Clustering Coefficient, were also calculated to provide a more complete characterisation of each player’s network role. While some degree of conceptual overlap may exist among these metrics,^{2,3,14,15} their combined presentation offers a multi-dimensional profile of individual involvement in the passing network.
2. Macro-level (collective): At the team level, Total Links, Network Density, Average Distance, Network Diameter, Network Heterogeneity, Global Centralisation, Global Prestige, Transitivity, Reciprocity, Assortativity Coefficient, and Network Transition Entropy were calculated to assess overall team cohesion, interaction diversity, and tactical adaptability. These metrics capture complementary aspects of collective behaviour: for example, Transitivity reflects the tendency to form passing “triangles” (three-player combinations) indicative of coordinated subgroups, while Global Centralisation and Global Prestige gauge how concentrated the playmaking is through particular individuals, and Network Transition Entropy quantifies the unpredictability of the team’s overall passing pattern.^{2,4–6,8,24,25} Collectively, they provide a comprehensive assessment of team-level network properties. (Mathematical definitions are provided in Supplemental Section S2).

This combination of metrics allows a multi-layered understanding of collective coordination and tactical functionality, linking structural positions within the network to functional and tactical aspects of coordinated performance. By including both traditional network indices and entropy-based measures, our SNA approach captures not only who the key players are,

but also the variability in how each of those players contributes to possession sequences.²

Markov-spectral modelling of passing networks

This section augments the SNA layer with a Markov-spectral model of ball circulation. Passing interactions are represented as a finite-state, time-homogeneous Markov chain on the active player set, providing compact indices of passing variability, diffusion speed, navigability, and structural robustness.^{9,18,37} By analysing the transition probabilities of passes as a stochastic process, we introduce a temporal dimension to the network analysis, enabling quantification of aspects like speed of ball circulation and distribution of possession that static metrics alone cannot capture.

For each match, a directed weighted adjacency matrix A was constructed, where each entry represents the number of passes from one player to another. From this adjacency matrix, a row-stochastic transition matrix P was derived on the active subgraph (players with at least one outgoing pass), where each entry represents the probability that a player passes to a specific teammate. When the empirical transition matrix was not irreducible, a teleportation correction ($\alpha = 0.15$) was applied to ensure irreducibility of the Markov chain. This value follows the standard parameter used in PageRank-based Markov processes, ensuring ergodicity while minimally perturbing the observed transition structure.^{37–39} All Markov-spectral quantities requiring a Stationary Distribution or spectral decomposition were computed from the ergodic chain.

From the transition matrix, we derived five families of Markov-spectral indices:

1. Long-run ball share and revisit rates: Stationary Distribution and Mean Recurrence Rime (MRT)^{18,40};
2. Passing uncertainty: Entropy Rate (bits/pass) and Perplexity^{41,42};
3. Diffusion and navigability: Spectral Gap, Kemeny’s Constant, and Mean First-Passage Times (MFPT)^{39,40,43};
4. Directional asymmetry: Asymmetry Index^{18,44};
5. Structural robustness and complexity: Algebraic Connectivity,^{45,46} von Neumann Entropy,⁴⁷ and Kirchhoff Index^{48,49} (Mathematical definitions are provided in Supplemental Section S3).

These indices characterise stochastic flow properties not captured by static SNA, providing a multidimensional profile of tactical functionality that complements structural metrics.

Integrated analytical framework: Metric summary and conceptual distinctions

Table 1 summarises the tactical interpretations of all SNA and Markov-spectral metrics used in this study,

Table 1. Summary of SNA and Markov-spectral metrics: tactical interpretation, status, and hypothesis linkage.

Domain	Level	Metric	Tactical interpretation	Status	Hypothesis
SNA	Micro	Degree centrality	Volume of direct involvement: Quantifies how frequently a player acts as a passer; indicating workload in possession circulation and immediate availability as an outlet; high values suggest greater involvement in ball circulation.	Primary	H1
SNA	Micro	Closeness centrality	Distribution speed: Assesses how quickly a player can reach (or be reached by) teammates; lower values suggest the ability to receive and distribute under time pressure; higher values may indicate peripheral positioning.	Primary	H1
SNA	Micro	Stress centrality	Transitional load: Identifies players who bear the highest volume of passing traffic between sectors, indicating responsibility in maintaining possession during phase transitions; high values indicate a key role in connecting team sectors.	Primary	H1
SNA	Micro	Betweenness centrality	Connective mediation: Identifies players who bridge different team sectors (defence-midfield-attack), enabling transitions and bypassing opposition pressure; high values indicate an essential role in team connectivity.	Primary	H1
SNA	Micro	Eccentricity centrality	Isolation risk: Identifies players who are most isolated in the tactical structure; high values suggest lower active participation and limited connectivity to the rest of the team.	Primary	H1
SNA	Micro	Degree prestige	Solicited player: Identifies players who are most sought out by teammates during the game, as measured by the number of passes received; high values indicate they are trusted targets in possession.	Primary	H1
SNA	Micro	Proximity prestige	Support availability: Analyses players who are fundamental for maintaining possession through close support; high values suggest reliability as a short-passing option.	Primary	H1
SNA	Micro	Eigenvector centrality	Influence through association: Measures a player's connectivity to other highly connected teammates; high values indicate involvement in the team's core passing combinations.	Primary	H1
SNA	Micro	Subgraph centrality	Local synergy: Evaluates participation in triangular passing combinations and small-group interactions; high values indicate contribution to coordinated sub-group play.	Primary	H1
SNA	Micro	PageRank centrality	Tactical reference: Identifies players who receive passes due to their perceived importance; high values suggest teammates actively seek them out as reliable options.	Primary	H1
SNA	Micro	Power centrality	Rhythm control: Assesses players who influence game tempo by controlling when and where passes are distributed; high values indicate strategic decision-making authority and tempo control.	Primary	H1
SNA	Micro	Centroid centrality	Spatial-organisational hub: Indicates the player best positioned to distribute play from a central network location; high values suggest tactical positioning and spatial awareness.	Primary	H1
SNA	Micro	Clustering coefficient	Local combination play: Identifies players who favour quick passing exchanges in tight spaces; high values indicate participation in short, repetitive passing patterns.	Primary	H1
SNA	Micro	Node transition entropy	Individual passing variability: Measures how evenly a player distributes passes among teammates. Distinction from centrality: A player may have high degree centrality (many passes) but low entropy (always passing to the same teammate—predictable), or moderate degree but high entropy (unpredictable distribution). Captures diversity of passing choices, not volume; high values indicate unpredictable passing distribution; low values suggest predictable patterns.	Exploratory	-

(continued)

Table 1. (continued)

Domain	Level	Metric	Tactical interpretation	Status	Hypothesis
SNA	Macro	Total links	Collective involvement: Total number of connected player pairs; reflects overall team participation in possession and interaction diversity; higher values indicate greater collective involvement.	Primary	H2
SNA	Macro	Network density	Interaction fluidity: Proportion of possible connections realised; higher density suggests quick passing options and supported play; lower density may indicate spacing or structured attacking patterns.	Primary	H2
SNA	Macro	Average distance	Structural proximity: Average path length between players; shorter distances indicate tight positioning and quick exchange potential; longer distances may suggest spacing or width.	Primary	H2
SNA	Macro	Network diameter	Maximum reorganisation length: Longest shortest path between any two players; indicates how quickly the team can reorganise possession across extremes of the formation; smaller values suggest faster reorganisation capacity.	Primary	H2
SNA	Macro	Network heterogeneity	Interaction concentration: Measures whether play is concentrated among a few players (high heterogeneity) or distributed more evenly (low heterogeneity). Distinction from centralisation: Heterogeneity reflects variance in involvement; centralisation reflects how much this variance is organised around specific individuals; high values indicate concentration around a few players; low values suggest balanced distribution.	Primary	H2
SNA	Macro	Transitivity	Triangulation tendency: Measures the proportion of triplet connections that form closed triangles; reflects capacity for quick passing combinations and small-group play; high values indicate strong small-group interaction capacity.	Primary	H2
SNA	Macro	Reciprocity	Bidirectional exchange: Shows balance in two-way passing between player pairs; high reciprocity suggests mutual trust and balanced distribution; low reciprocity indicates directed, progression-oriented play.	Primary	H2
SNA	Macro	Global centralisation	Hub dependence: Indicates whether the team relies on one or a few key players to conduct play; higher values suggest vulnerability if central players are marked out.	Primary	H2
SNA	Macro	Global prestige	Attraction to key players: Identifies players who receive passes due to technical ability or positioning; high values suggest targeted progression through specific individuals.	Primary	H2
SNA	Macro	Assortativity coefficient	Role-based connectivity: Identifies whether players tend to connect with similar (positive) or different (negative) roles; negative values suggest functional integration across positional lines; positive values indicate within-role clustering.	Primary	H2
SNA	Macro	Network transition entropy	Team-level passing unpredictability: Global measure of how evenly passes are distributed across all possible connections. Distinction from density: Density measures how many connections exist; entropy measures how evenly those connections are used. A dense network can be predictable (low entropy) if passes concentrate on a few links; high values indicate greater unpredictability for the opposition.	Exploratory	-
Markov	-	Stationary distribution (π)	Long-run possession share: Probability of finding the ball with a specific player after many passes. Distinction from degree centrality: Degree counts total passes; the stationary distribution accounts for flow, who controls the ball in steady-state dynamics, weighted by who passes to them; higher values indicate greater long-run possession share.	Exploratory	H3

(continued)

Table 1. (continued)

Domain	Level	Metric	Tactical interpretation	Status	Hypothesis
Markov	–	Mean recurrence time (MRT)	Recycling speed: Expected number of passes for the ball to return to a specific player; lower values indicate quicker ball recovery/recycling through that player; quantifies how effectively a player resets possession.	Exploratory	H3
Markov	–	Entropy rate	Average uncertainty per pass: Team-level, possession-weighted measure of decision unpredictability. Integrates individual passing entropies weighted by each player's possession share. Distinction from node entropy: Node entropy measures individual variability; entropy rate captures how this variability propagates through actual possession sequences; higher values indicate greater tactical unpredictability.	Exploratory	H3
Markov	–	Perplexity	Effective options per action: Scale-free transformation of entropy rate; interprets unpredictability as the equivalent number of equally likely passing options per action; higher values indicate more diverse attacking possibilities.	Exploratory	H3
Markov	–	Spectral gap	Possession diffusion speed: Quantifies how quickly the ball circulates through the network. Distinction from density/closeness: Density and closeness measure structural connectivity; spectral gap measures dynamic mixing—the rate at which possession can move from any player to any other through repeated passes; higher values indicate faster possession circulation.	Exploratory	H3
Markov	–	Kemeny's constant	Network navigability: Expected number of passes to reach a randomly chosen teammate, regardless of starting point. Distinction from average distance: Average distance counts shortest paths; Kemeny's constant reflects expected traversal time in the actual stochastic process of play, accounting for passing probabilities; lower values indicate more efficient network navigation.	Exploratory	H3
Markov	–	Mean first-passage time (MFPT)	Target accessibility: Expected number of passes for the ball to reach a specific player (e.g. striker) from any starting point. Quantifies build-up efficiency to key attacking roles; lower values indicate more efficient build-up to targeted players.	Exploratory	H3
Markov	–	Asymmetry index	Directional bias: Degree of preferential flow in possession circulation; low values indicate balanced left-right distribution; high values suggest attacking down a predominant side.	Exploratory	H3
Markov	–	Algebraic connectivity	Structural robustness: Resistance to network fragmentation; higher values indicate the ability to maintain cohesion under pressure and adapt to player removal (e.g. tactical substitutions).	Exploratory	H3
Markov	–	von Neumann entropy	Topological complexity: Diversity of passing configurations in the network; higher values suggest multiple structural pathways and adaptive reconfiguration capacity.	Exploratory	H3
Markov	–	Kirchhoff Index	Path redundancy: Global effective resistance; lower values indicate multiple alternative routes for possession progression, reducing predictability and increasing resilience to pressing.	Exploratory	H3

Note. Distinguishing centrality from entropy: Centrality metrics (Degree, Betweenness, and Eigenvector) quantify how much a player participates (volume of involvement). Entropy-based metrics (Node Transition Entropy and Entropy Rate) quantify how variably they participate (diversity of passing choices). A player can have high centrality but low entropy (predictable hub) or moderate centrality but high entropy (unpredictable distributor). Both dimensions matter for tactical profiling.

Distinguishing structural from flow metrics: SNA metrics (Density, Transitivity, and Assortativity) describe the static architecture of who-connects-to-whom. Markov-spectral metrics (Spectral Gap, Kemeny's Constant, and MFPT) describe stochastic flow properties, how possession actually moves through this architecture over time. A network can have an identical structure yet exhibit markedly different flow dynamics depending on how passing probabilities are distributed.

Tactical interpretations for SNA metrics are based on Refs. ^{1–6,24,25,30,34,35} and interpretations for Markov-spectral metrics draw on Refs. ^{18,18,37–49}

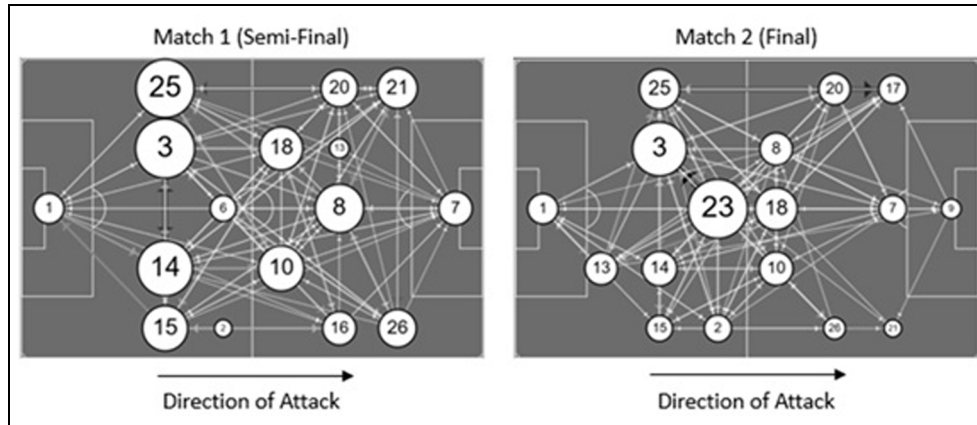


Figure 2. Representation of interactions among players of the Portuguese National Football Team in each match (semi-final and final) of the final stage of the 2025 Nations League (network generated using uPATO®).

Node size reflects relative interaction frequency; edge thickness represents pass frequency between players; players are positioned by tactical role to illustrate formation structure; and numerical values for interactions are provided in the text.

along with their status (primary vs exploratory) in relation to the research hypotheses. All SNA metrics are considered primary, forming the established framework for analysing football passing networks. Markov-spectral metrics are exploratory, representing the novel contribution of this study. It is important to note that entropy-based quantities across the two analytical layers are complementary rather than numerically interchangeable. In the SNA layer, Node Transition Entropy characterises the variability of an individual player's pass distribution within the uPATO® framework, whereas in the Markov-spectral layer the entropy rate quantifies possession-weighted uncertainty of the stochastic process on the active player network. Accordingly, the former is a player-level distributional descriptor, while the latter is a process-level flow descriptor.

Results

The following sections present the results in detail, including: (i) match networks and tactical organisation, (ii) descriptive SNA comprising micro-level analysis of individual player performance and macro-level network metrics capturing collective team behaviour, and (iii) Markov-spectral analysis quantifying uncertainty, diffusion, navigability, and robustness.

Networks and tactical organisation

Figure 2 illustrates the network of interactions among players of the Portuguese National Football Team in each match (semi-final and final) of the 2025 Nations League.

In Match 1, the team prioritised midfield control and rapid transitions both down the wings and through the centre, employing a 1-4-2-1-3 formation: a goalkeeper; four defenders (two central defenders and two full-backs (left-back and right-back)); two deep-lying

central midfielders responsible for defensive protection and ball distribution; an attacking midfielder linking midfield to attack; and three forwards (two wingers and a central forward).

It was observed that Player 3 (central defender) recorded the highest number of interactions, with 90 passes (50 made and 40 received), followed by Player 25 (left back) with 86 interactions (40 made and 46 received), and Player 14 (central defender) with 80 interactions (42 made and 38 received).

In Match 2, the team employed a different formation, using a 1-4-1-2-3 formation: a goalkeeper; four defenders (two central defenders and two full-backs (left-back and right-back)); a deep-lying defensive midfielder shielding the backline; two advanced central midfielders linking play and enhancing offensive presence; and three forwards (two wingers and a central forward).

The players with the highest involvement were Player 23 (central midfielder) with 127 interactions (70 made and 57 received), Player 3 (central defender) with 110 interactions (56 made and 54 received), and Player 18 (central midfielder) with 77 interactions (39 made and 38 received). Across both matches, the highest total interactions were recorded for Player 3 (200), Player 23 (175), and Player 25 (155).

Descriptive social network analysis results

Figure 3 presents the micro-level analysis of individual player performance using centrality metrics, identifying players with the highest involvement in ball build-up, circulation, and possession maintenance.

In both matches, the data suggested that Player 3 (central defender), Player 23 (central midfielder), and Player 25 (left back) figured as central players within the passing network.

Degree, Closeness, Betweenness, and Eigenvector Centralities consistently positioned Player 3 (central

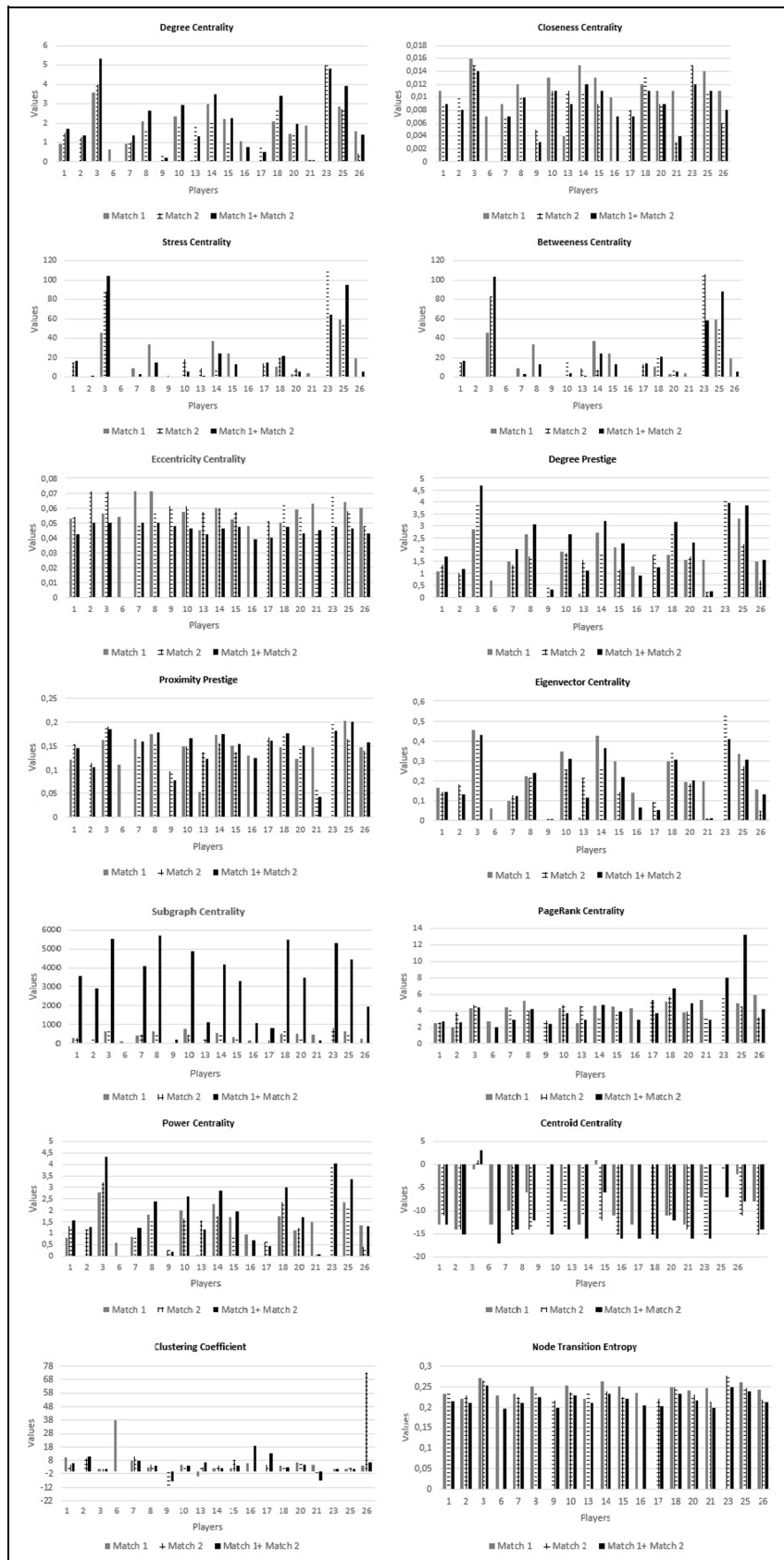


Figure 3. Analysis of individual player performance using mathematical models (micro-analysis). Bar heights represent metric values for each player; numerical values are reported in the text.

defender) at the core of interactions. Player 23 (central midfielder) and Player 25 (left back) also demonstrated high connectivity. Eccentricity, Degree Prestige, Proximity Prestige, PageRank, Power, and Centroid Centralities further reinforced the prominence of these three players as structural hubs. Stress and Subgraph Centralities further indicated that these players participated in alternative pathways and local network structures.

Clustering Coefficient values suggested interactions on the wide areas, with wingers and full-backs providing lateral support. Furthermore, Node Transition Entropy revealed that Player 3 (central defender) exhibited the greatest variability in passing, followed closely by Player 23 (central midfielder) and Player 25 (left back).

Figure 4 illustrates collective team performance, as captured by metrics such as Total Links, Network Density, Average Distance, Network Diameter, Network Heterogeneity, Global Centralisation, Global Prestige, Transitivity, Reciprocity, Assortativity Coefficient, and Network Transition Entropy.

The analysis of collective performance revealed consistent patterns of interaction among players, alongside some variations between matches. The total number of connections (Total Links) increased from 373 in Match 1 to 414 in Match 2, yielding 787 interactions overall. Network Density remained relatively stable (0.111 in Match 1; 0.109 in Match 2; total = 0.115).

Average Distance decreased from 9.125 to 7.462 (total = 9.324). Network Diameter increased from 22.167 to 23.750 (total = 25.333). Network Heterogeneity rose from 0.598 to 0.725 (total = 0.737).

Transitivity decreased from 6.303 to 5.094 (total = 5.783). Reciprocity increased slightly from 0.649 to 0.662 (total = 0.727). Global Centralisation increased from 0.145 to 0.231 (total = 0.200). Global Prestige remained relatively stable (0.134–0.145; total = 0.140).

The Assortativity Coefficient was negative in both matches (−0.046 and −0.136; total = −0.132). Moreover, Network Transition Entropy increased from 3.894 to 3.989 (total = 4.157).

Markov-spectral results

Complementing the descriptive network results reported previously, the Markov-spectral layer provides compact indicators of the stochastic flow properties of possession (e.g. uncertainty, diffusion, navigability, and robustness) averaged across each match. Table 2 summarises team-level values by match, and Table 3 presents the top-five players by Stationary Distribution per match, where we additionally report the π -weighted MFPT to the centre-forward (player 7). It is noteworthy that the “Total” network was obtained by summing the two match-specific adjacency matrices before recalculating all indices; it should be interpreted as a pooled descriptive representation across the two matches, not as a temporal trend or inferential benchmark.

Entropy Rate increased from 2.772 bits in Match 1 to 2.839 bits in Match 2 and was highest in the aggregated network (3.126 bits), indicating broader effective choice (Perplexity) per pass. The Spectral Gap rose from 0.454 in Match 1 to 0.530 in Match 2, consistent with faster diffusion across the network structure. These observed differences suggested that possession spread more rapidly and evenly through the team’s network in Match 2, which may reflect more fluid ball circulation. Kemeny’s Constant was higher in Match 2 (15.73) than in Match 1 (14.12), suggesting more distributed exploration prior to reaching a random teammate. This finding could indicate the ball traversed a wider range of players before reaching an arbitrary teammate in Match 2, potentially reflecting a more deliberate and patient build-up of play. Asymmetry decreased from 0.354 in Match 1 to 0.320 in Match 2, suggesting a more balanced directional circulation of possession. This observation may imply that the team was less reliant on attacking in a single predominant direction in Match 2. Algebraic Connectivity increased from 0.435 to 0.583, and von Neumann Entropy increased from 3.549 bits to 3.647 bits, suggesting improved robustness and topological complexity. These changes could indicate that the second match’s passing network was structurally more cohesive and exhibited greater complexity in its configuration. The Kirchhoff Index rose from 18.81 in Match 1 to 23.13 in Match 2, but was lowest in the aggregate (15.87), suggesting that local bottlenecks may have eased while global effective resistance increased in the single-match setting.

A direct comparison between the insights from traditional SNA and those from the Markov-spectral layer illustrates their complementary value. While SNA successfully identified the stable structural hubs (players 3, 23, and 25) and described macro-level cohesion (e.g. stable network density), the Markov-spectral indices quantified aspects of ball circulation that were not captured by structural metrics alone. Specifically, the increase in spectral gap (0.454–0.530) revealed faster ball diffusion in Match 2, a nuance of fluidity not discernible from static connectivity. Concurrently, the rise in Kemeny’s Constant (14.12–15.73) and Mean First-Passage Time to the centre-forward (15.6–22.2 passes) quantified a more patient, exploratory buildup, despite SNA indicating greater hub concentration (increased heterogeneity and centralisation). Furthermore, the increase in Entropy Rate (2.772–2.839 bits/pass) provided a model-based measure of per-action decision uncertainty, complementing SNA’s node-level entropy. Together, these observations suggest that the Markov-spectral framework captures stochastic flow properties of possession that complement the structural insights provided by SNA.

The Stationary Distribution presented in Table 2 aligns with the descriptive centralities reported earlier. Note that all values are reported per match and for the aggregated network, providing a quantitative overview

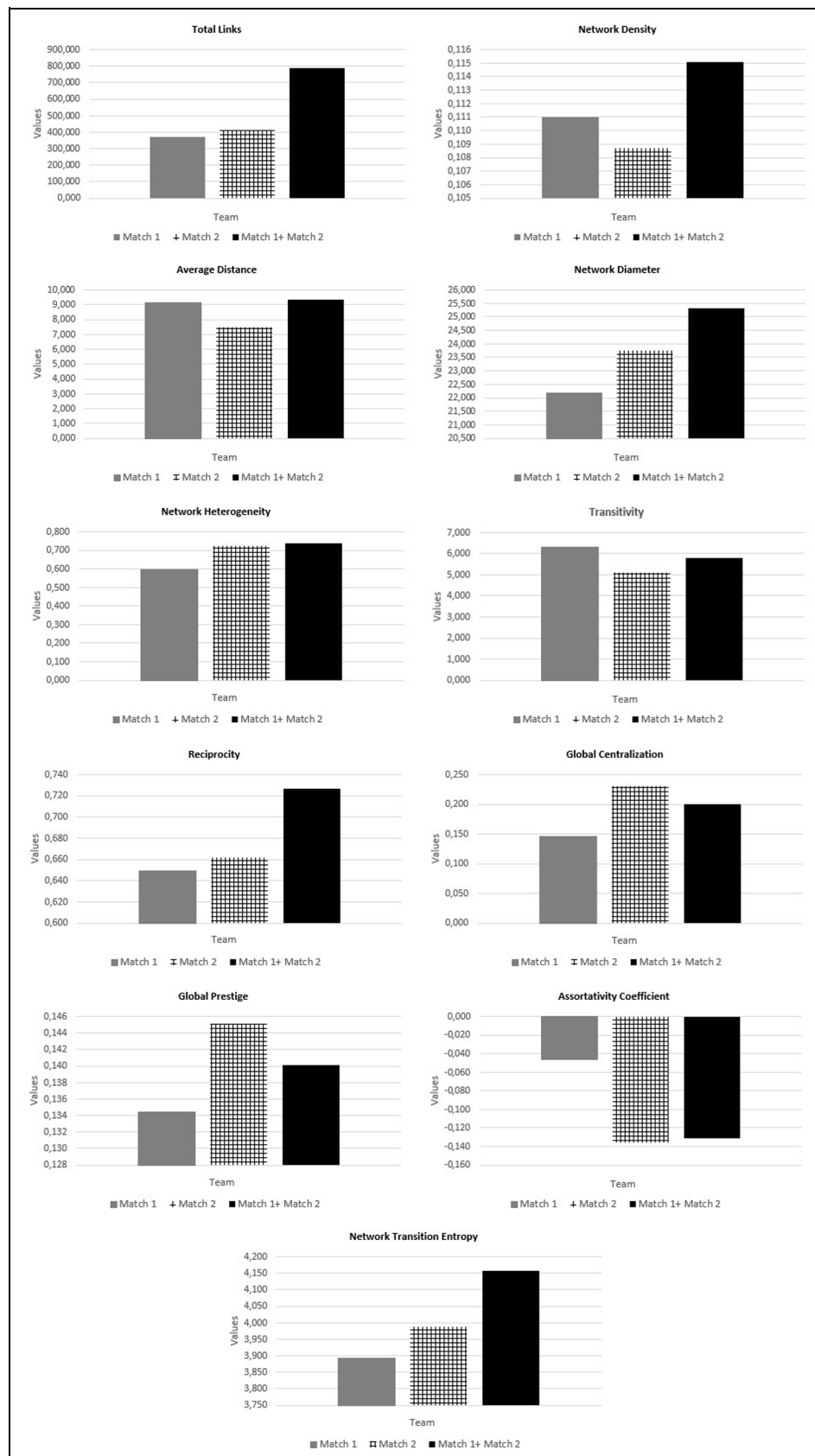


Figure 4. Analysis of the team's collective performance using mathematical models (macro-analysis). Bar heights represent metric values for the team; numerical values are reported in the text.

Table 2. Team-level Markov-spectral metrics per match.

Match	Entropy rate (bits)	Perplexity	Spectral gap	Kemeny's constant	Asymmetry index	Algebraic connectivity	von Neumann entropy (bits)	Kirchhoff Index
1	2.772	6.83	0.454	14.12	0.354	0.435	3.549	18.81
2	2.839	7.16	0.530	15.73	0.320	0.583	3.647	23.13
Total	3.126	8.73	0.663	16.66	0.274	0.636	3.739	15.87

↑ Entropy Rate: more unpredictable passing; ↑ Perplexity: more effective passing options; ↑ Spectral Gap: faster ball circulation; ↑ Kemeny's Constant: more patient build-up; ↓ Asymmetry Index: more balanced left-right play; ↑ Algebraic Connectivity: stronger structural cohesion; ↑ von Neumann Entropy: higher topological complexity; ↓ Kirchhoff Index: greater global path redundancy.

Table 3. Top-five by stationary distribution (π): match 1 (left) versus match 2 (right).

Match 1			Match 2		
Player	Position	π	Player	Position	π
25	Left back	0.131	23	Central midfielder	0.140
8	Central midfielder	0.107	3	Central defender	0.125
14	Central defender	0.098	25	Left back	0.098
3	Central defender	0.096	18	Central midfielder	0.090
10	Central midfielder	0.072	10	Central midfielder	0.064
π -weighted MFPT to the centre-forward (Player 7) in match 1: 15.6 passes.			π -weighted MFPT to the centre-forward (Player 7) in match 2: 22.2 passes.		

Stationary Distribution (π) = Long-run possession share (higher values = primary circulation hubs). π -weighted MFPT to centre-forward = expected passes to reach the centre-forward; ↑ from 15.6 to 22.2 reflects more patient build-up in Match 2.

of team-level ball circulation properties and individual possession shares. Table 3 confirms that Player 25 (left back) and Player 23 (central midfielder) hosted a large share of long-run possession. The Mean First-Passage Time result quantifies the expected number of passes needed for the ball to reach the centre-forward (player 7), providing a functional complement to degree or betweenness-based descriptions. This result means that, in Match 1, on average about 15.6 passes were required for the team to work the ball to the centre-forward, whereas in Match 2 this expected length increased to 22.2 passes, highlighting the shift towards a more patient, control-oriented buildup in the second match.

Discussion

This study employed a multi-method approach, integrating SNA with Markov-spectral techniques, to characterise the structural and stochastic flow properties of the Portuguese National Team's passing networks during the 2025 Nations League finals.

The SNA results underscore the role of a central defender (player 3) as the main network hub, exhibiting high Degree, Betweenness, Eigenvector, and Stress Centrality, consistent with previous studies indicating that defenders can function as critical "hubs" during build-up phases in possession-oriented teams.^{1,2} This observation aligns with findings that football passing networks often exhibit scale-free properties with

emergent key players acting as hubs.⁵⁰ Player 3's highest Centroid Centrality value (+ 3.000) identifies him as a central organiser within the network, a role similarly observed in elite UEFA Champions League teams where defenders were decisive for team connectivity.²⁶ On the other hand, the concurrent prominence of a central midfielder (player 23) and a left-back (player 25) aligns with the literature highlighting midfielders' critical contribution to offensive transitions and full-backs' role in providing width and creating scoring opportunities.^{7,8,51} The Node Transition Entropy value for Player 3 was the highest among teammates, reflecting greater individual passing variability, which has been associated with increased unpredictability in ball distribution.^{24,25} This structural pattern, where cohesive networks with central defenders and midfielders as key connectors are linked to successful performance, is supported by recent syntheses of the literature.^{2,3,15} The presence of multiple key players (player 3, player 23, and player 25) reflects the concept of system redundancy,^{2,52} a hallmark of adaptive complex systems where functionality may be maintained even if a key player's influence is reduced. Such redundancy could potentially enhance the team's robustness, as alternative playmakers might assume control if one hub is neutralised. Collectively, these structural findings are consistent with our first hypothesis (**H1**), indicating that SNA can identify stable structural hubs, namely, the central defender (player 3), central midfielder (player 23), and left-back (player 25), who consistently

act as central conduits for ball progression across both matches.

At the macro level, collective performance metrics indicate a balance between structural stability and observed variations. Network Density remained relatively stable (~ 0.115) across matches, consistent with cohesion despite tactical variations.²⁶ However, increases in Network Heterogeneity and Global Centralisation from Match 1 to Match 2 indicate a growing concentration of interactions around key players, which may reflect enhanced offensive orchestration while potentially signalling vulnerability if central nodes are neutralised, an insight relevant for opposition scouting and tactical preparation.²⁹ Additionally, the negative Assortativity Coefficient is consistent with players tending to connect with teammates of differing characteristics, which may promote functional diversity and integration within the network, analogous to the interaction of specialised modules in complex adaptive systems.^{1,4,52} The rise in Network Transition Entropy reflects increasing unpredictability in passing sequences, which has been associated with making the team more difficult to defend against.^{24,25} Formation adjustments from 4-2-1-3 to 4-1-2-3 shifted playmaking responsibility, as evidenced by the increased involvement of Player 23 in Match 2, illustrating the context-dependent nature of centrality and the adaptive redistribution of influence across the team.^{7,8,53} Collectively, these findings from the SNA layer illustrate the capacity of this approach to provide a multi-dimensional assessment of network properties, offering actionable insights for evaluating team connectivity, identifying structural characteristics.⁵⁴ These results relate to our first research question.

While prior research has successfully established the Markov-chain model as a valid approximation for the growth of passing networks over time,¹⁶ the present findings suggest that this framework's potential extends far beyond descriptive network evolution.

Complementing these structural insights, the Markov-spectral layer frames passing as a stochastic diffusion process on the active player set, yielding interpretable quantities that complement classical SNA. This integrated perspective, which models passing as a stochastic process directly on the player network, aligns with, and substantially extends, recent approaches using Markovian models to unravel patterns in football passing sequences.¹¹ Moving beyond the examination of network growth,¹⁶ the proposed model provides a nuanced view of the quality of ball circulation through metrics such as Entropy Rate (bits per pass) which captures uncertainty in pass destination,^{41,42} Spectral Gap, Kemeny's Constant measuring network navigability,^{39,40,43} Stationary Distribution, MFPTs, Algebraic Connectivity assessing structural robustness,^{45,46} von Neumann Entropy quantifying topological complexity,⁴⁷ and Kirchhoff Index evaluating global effective resistance.^{48,49}

A critical contribution of this study is the demonstration of Markov-spectral metrics revealing what

static SNA cannot. While SNA successfully identified stable structural hubs and described macro-level cohesion, these metrics are inherently static, describing the aggregate pattern of who connected to whom, but not how possession flowed through time. The Markov-spectral layer addresses distinct functional questions that SNA alone cannot answer. First, flow dynamics: the Spectral Gap quantifies the speed of possession diffusion through the network, a property invisible to static SNA. Second, navigability and build-up patience: Kemeny's Constant and MFPT reveal how efficiently possession reaches teammates and attacking targets, complementing centrality measures that only identify who is involved. Third, unpredictability at the team level: the Entropy Rate provides a possession-weighted measure of decision uncertainty per action, transcending aggregated individual behaviours. Fourth, robustness and resilience: Algebraic Connectivity and von Neumann Entropy quantify structural cohesion and topological complexity, properties related to the team's ability to maintain organisation under pressure. These dimensions are conceptually orthogonal to structural metrics and together provide a multi-layered functional profile of team behaviour that static analysis alone cannot achieve, capturing the stochastic flow properties of possession.

This approach aligns with and extends recent research applying Markov models to football, such as the work of Liu et al.,¹⁷ who used state transition matrices between field zones to identify general and threatening attacking patterns in the 2018 World Cup. While their study effectively captured the spatial dynamics of team attacks (the "where"), our Markov-spectral analysis of the player network elucidates the underlying relational and structural dynamics (the "how" and "through whom"), offering a complementary and multi-layered understanding of tactical functionality.

In the present dataset, higher Entropy Rate (2.772–2.839 bits/pass) and Spectral Gap (0.454–0.530) in Match 2 indicate more unpredictable and faster-circulating possession in the final, consistent with enhanced tactical adaptability.²⁵ The increase in Kemeny's Constant (14.12–15.73) and MFPT to the centre-forward (15.6–22.2 passes) reflects more patient, exploratory build-up, with possession traversing the network more thoroughly before reaching attacking targets. Concurrent rises in Algebraic Connectivity (0.435–0.583) and von Neumann Entropy (3.549–3.647 bits) denote enhanced structural cohesion and topological complexity, suggesting greater resilience under pressure.¹¹ The decline in Asymmetry (0.354–0.320) indicates more balanced left-right circulation, while the lowest Kirchhoff Index in the aggregated network (15.87) confirms global path redundancy. These observed differences between matches, characterised by preserved network cohesion alongside increased Entropy Rate, Spectral Gap, and Kemeny's Constant, are consistent with our second hypothesis, showing

measurable variations in stochastic flow properties while maintaining structural integrity. Elevated Kemeny's constant in Match 2 is consistent with possession exploring the network more evenly prior to reaching a random teammate, which may reflect a deliberate, patient build-up approach, similar to patterns observed in successful possession-based teams.³⁰ Similarly, the π -weighted MFPT to the centre-forward (player 7) increased from approximately 15.6 to 22.2 passes, quantifying a shift towards controlled recycling through midfield zones before advancing to the centre-forward, a strategic pattern that contrasts with more direct attacking approaches reported in other elite teams.² Taken together with the observed increases in connectivity and complexity metrics, as well as the lowest Kirchhoff Index in the aggregated network, these results suggest a robust and efficient network structure that aligns with structural properties linked to successful performance in elite football.⁴ Analysis of the Stationary Distribution (π) confirmed that the left-back (player 25) and central midfielder (player 23) served as primary possession hubs, corroborating SNA findings and underscoring their crucial playmaking roles within the team's tactical framework. The integration of π with MFPT provides a functional perspective on player contributions, quantifying both their centrality and the expected number of interactions required to advance the ball to key nodes. Specifically, the analysis revealed that reaching the centre-forward (player 7) required an average of 15.6 passes in Match 1, increasing to 22.2 passes in Match 2. This approach extends the evaluation of offensive progression beyond conventional metrics,^{4-6,52} offering a perspective of possession efficiency and observed differences between matches. Crucially, the Markov-spectral indices provided insights that complement and extend beyond traditional SNA metrics, consistent with our third hypothesis.

Thus, this integrated analysis illustrates that elite football passing networks can be comprehensively characterised through the combination of structural (SNA) and stochastic flow perspectives (Markov-spectral). The framework successfully captured both invariant hub structures and match-specific flow properties, offering a multi-layered profile of team behaviour. These findings support the utility of combining these methodological approaches for performance analysis in context, providing practitioners with insights into both organisational stability and functional dynamics.

Limitations

Despite the insights provided by the integrated SNA and Markov-spectral framework, several limitations should be acknowledged.

First, the study focusses on a single team across only two matches of a specific tournament, which inherently limits the generalisability of the findings and prevents the detection of consistent patterns that might emerge

over a larger sequence of matches. As an exploratory proof-of-concept, the sample size is appropriate for demonstrating feasibility, but statistical power is insufficient for inferential testing.⁴¹ Given the exploratory proof-of-concept design and the limited number of matches analysed ($n = 2$), inferential procedures such as bootstrap confidence intervals or permutation tests would not provide stable or meaningful uncertainty estimates. Consequently, the present study focusses on descriptive indices, while future research with larger datasets should implement resampling-based uncertainty quantification. Critically, with only two matches analysed, it is not possible to determine whether the observed variations between matches reflect genuine tactical adaptations by the team in response to different opponents and match contexts, or whether they represent normal stochastic fluctuations inherent to passing sequences in football. The observed differences in Entropy Rate, Spectral Gap, and other indices, while descriptively interesting, cannot be statistically distinguished from random match-to-match variability. This fundamental ambiguity underscores the exploratory nature of this study and highlights the necessity of replicating this analysis across a larger number of matches to establish whether such variations are systematic and meaningful, or merely incidental. Furthermore, given that passing networks inherently possess structure, the framework's discriminative power must be critically evaluated. Future studies should test whether these metrics can reliably distinguish between distinct playing styles (e.g. possession-based vs counter-attacking), match outcomes (win/draw/loss), or performance phases (e.g. high- vs low-pressure moments). It is possible that certain network configurations may appear similar across different tactical contexts, underscoring the need for comparative validation. Furthermore, the discriminative power of the proposed metrics across different tactical styles (e.g. counter-attacking vs possession-based teams) remains untested. Future validation should examine whether Entropy Rate, Spectral Gap or Kemeny's Constant can distinguish between playing philosophies more effectively than traditional SNA metrics.

Second, the analysis is based on aggregated passing networks over entire matches, which may obscure temporal fluctuations and higher-order sequences that influence tactical decisions. Moreover, the Markov layer should be interpreted as a parsimonious first-order approximation of possession flow rather than as an empirically validated generative model of passing sequences. Future work should therefore test higher-order dependence and temporal non-stationarity using event-level ordered data. The aggregation across full matches limits our ability to capture within-match dynamics and real-time tactical adaptations. Consequently, interpretations regarding observed between-match variations should be understood as describing average match properties rather than moment-to-moment strategic shifts. In particular, we

did not capture within-match dynamics; applying a time-resolved or sliding window network analysis could reveal how the team's connectivity and key indices change during different phases (e.g. before vs after a goal or as fatigue sets in).

Third, the current model does not explicitly incorporate spatial coordinates or player movement trajectories, which could further elucidate how positional dynamics interact with passing structure. Incorporating spatiotemporal data, such as mapping passes to specific field zones, would enable analysis of how network measures relate to formation and space management on the field. Moreover, like much of the existing literature, our focus was on offensive passing networks; future research should extend this framework to defensive interactions and transition plays, constructing analogous defensive networks to achieve a more comprehensive evaluation of team performance.

Future research

Future research should aim to extend this framework by incorporating time-resolved network dynamics, allowing for the detection of transient motifs, momentum shifts, and situational patterns. Concretely, this approach could involve applying sliding-window or phase-based analyses to examine how Markov-spectral indices fluctuate during key match periods (e.g. after conceding a goal or during high-pressing phases). Longitudinal studies tracking network and spectral evolution across an entire tournament or season would allow for a better understanding of how observed variations unfold over time and under varying competitive pressures.

Integrating spatiotemporal data would also enable the examination of the interaction between network structure and positional play, enhancing the link between structural metrics and functional outcomes. Additionally, applying this methodology across multiple teams, leagues, and international competitions could provide comparative insights into different tactical systems. Such comparative work is essential to establish whether the observed patterns are generalisable or context-dependent, and to build normative references for key metrics across styles and levels of play. Future studies should also employ null models and random network comparisons to establish baselines for spectral gap, entropy rate, and other indices, enabling more robust interpretation of observed values.^{45,46} Expanding the analyses to underrepresented domains, such as women's football or youth competitions, would also test the framework's applicability across varying match contexts and address identified gaps in network analysis research. Beyond descriptive comparisons, future studies should also test the predictive validity of the proposed indices, for example, by examining whether Entropy Rate, Spectral Gap, or Kemeny's Constant can forecast match outcomes or offensive

effectiveness more accurately than traditional SNA metrics alone.

Ultimately, such efforts will help establish the framework's versatility and facilitate the development of even more sophisticated models (e.g. machine learning approaches) to predict performance or optimise team strategies based on network-derived indicators.

Future applications of the integrated framework to larger datasets will enable robustness assessment through resampling techniques (e.g. bootstrap and permutation tests) and systematic sensitivity analyses of key parameters (e.g. teleportation correction). This framework will help establish whether the observed between-match variations reflect consistent tactical adaptations or fall within expected stochastic fluctuations.

Practical implications

The integrated SNA and Markov-spectral framework introduced in this study offers several potential insights for football practitioners, performance analysts, and coaching staff. The methodological approach moves beyond descriptive analytics, providing stochastic flow indicators that may inform training design, tactical preparation, and post-match analysis. Coaching staff could employ Entropy Rate and Spectral Gap to monitor passing unpredictability and diffusion speed, delivering objective measures of aspects related to tactical flexibility and ball circulation efficiency.

Augmenting SNA with a Markov-spectral model provides a concise, model-based perspective on unpredictability, fluidity, navigability, and robustness of team play. These metrics are straightforward to compute from standard adjacency matrices yet appear sensitive to tactical reorganisation between matches. Staff could track Entropy Rate and Spectral Gap to target decision-making variety and diffusion speed, use MFPTs to shorten the time to finishing roles, and monitor algebraic connectivity and Kirchhoff Index to evaluate resilience under opponent pressure or personnel changes.

From a strategic perspective, Kemeny's Constant and MFPT quantify network navigability and attacking progression. For instance, the MFPT to the centre-forward indicates how efficiently possession reaches key attacking players, supporting the evaluation of build-up effectiveness against specific match plans. Such indices may enable staff to tailor pre-match preparations and in-match strategies to address offensive fluidity.

Methodologically, this study contributes a reproducible analytical framework that bridges structural and stochastic flow perspectives, allowing researchers and practitioners to link network topology directly to functional outcomes. It offers a potential decision-support tool for coaches, technical staff, performance analysts, and researchers, enabling the identification of key players, the assessment of offensive efficiency, and the evaluation of observed tactical variations. Moreover,

Algebraic Connectivity and the Kirchhoff Index offer insight into structural robustness, highlighting dependencies within the passing network that could be exploited by opponents. These measures could inform both defensive organisation and opposition analysis, helping to identify potential areas of interest and reinforce tactical resilience. Together, these metrics form a potential toolkit for performance enhancement, supporting post-match analysis and the optimisation of training processes to improve collective performance.

It is important to note that these practical implications are proposed based on the exploratory findings of this study. Further validation across larger datasets and competitive contexts is necessary before these metrics can be reliably integrated into real-time decision-making processes.

Conclusion

Across both matches, SNA identified a consistent core of highly connected players, namely a central defender, a central midfielder, and a left-back, who accumulated the highest levels of interaction and occupied central positions within the passing network. At the team level, Network Density remained relatively stable between matches, while Network Heterogeneity and Global Centralisation increased in the final, indicating a greater concentration of interactions around key players.

Markov-spectral analysis revealed measurable differences between matches in several spectral indices. The Entropy Rate and corresponding Perplexity were higher in the final, reflecting a larger effective set of passing options per action. The Spectral Gap increased from the semi-final to the final, reflecting faster diffusion of possession across the network. Kemeny's Constant and the π -weighted Mean First-Passage Time (MFPT) to the centre-forward were also higher in the final, indicating longer exploratory passing sequences before reaching attacking targets. Directional Asymmetry decreased, while Algebraic Connectivity and von Neumann Entropy increased, describing changes in circulation balance and network structure between matches. When both matches were aggregated, the resulting network exhibited the highest Entropy Rate, Spectral Gap, and Algebraic Connectivity, alongside the lowest Kirchhoff Index, summarising the global structural and stochastic flow properties of the team's passing behaviour across the tournament's final stage.

Overall, the results document stable hub structures at the individual level, alongside match-to-match variation in stochastic flow, diffusion speed, and navigability of the passing network under high competitive pressure. Collectively, this study provides a methodological proof-of-concept for the integrated SNA and Markov-spectral framework, demonstrating how combining structural and stochastic flow metrics can reveal observed differences, such as increased fluidity, patience, and unpredictability, that are not apparent from either

perspective alone. The approach moves beyond describing network topology to quantifying functional flow, thereby offering a multi-layered toolkit for performance analysis. It is important to emphasise that these findings are context-specific, derived from an exploratory case study of two matches, and should not be interpreted as generalisable principles of elite football behaviour. The patterns observed herein are specific to this particular team, tournament, and competitive context, and require validation across larger samples, multiple teams, and diverse competitive settings before any broader generalisations can be considered. Further research with expanded datasets is necessary to determine whether the observed between-match variations reflect consistent tactical adaptations or are within the range of normal match-to-match fluctuation.

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Ethical considerations

This study was approved by the Ethics Committee of the Faculty of Sport Sciences and Physical Education, University of Coimbra (reference number CE/FCDEF-UC/0013/2025).

Consent to participate

Informed consent was not required for this study, as it was based on the analysis of anonymised performance data obtained from Wyscout[®], a commercial football performance analysis platform.

Author contributions

Conceptualisation, J.G., M.S.C., and G.D.; Methodology, J.G. and M.S.C.; Software, J.G. and M.S.C.; Validation, J.G., M.S.C., and G.D.; Formal

analysis, J.G., M.S.C., and G.D.; Investigation, J.G., M.S.C., R.M., and G.D.; Resources, J.G., M.S.C., and G.D.; Data curation, J.G., M.S.C., R.M., and G.D.; Writing – original draft preparation, J.G., M.S.C., and G.D.; Writing – review and editing, J.G., M.S.C., G.D., R.S.M., and V.V.; Visualisation, J.G., M.S.C., G.D., R.S.M., and V.V.; Supervision, G.D., R.S.M., and V.V.

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Declaration of conflicting interests

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Data availability statement

The data that support the findings of this study are available from Wyscout®, a commercial football performance analysis platform. Due to licensing restrictions, these data are not publicly available. Access to the data may be granted upon request, subject to Wyscout® permission. No persistent identifier (DOI) is available for these data.

Code availability statement

The code used for statistical analyses in this study is not publicly available. Requests for access to the analysis scripts may be directed to the corresponding author.

Supplemental material

Supplemental material for this article is available online.

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